

Topology First-Year Exam, August 2019, Part 1

Work all problems.

Theorems and Proofs.

In this section state and prove each theorem. The proofs should be complete without being tedious.

1. Let X be a metric space and suppose that $A_\lambda \subset X$ is connected for all $\lambda \in \Lambda$. Suppose that there is an $x_0 \in X$ such that $x_0 \in A_\lambda$ for all λ . Show that $\bigcup_{\lambda \in \Lambda} A_\lambda$ is connected.
2. State and prove the Contraction Mapping Theorem.
3. Let X be a Hausdorff space. Show that if $A \subset X$ is compact, then A is closed.
4. Let X be a topological space. Suppose that $f : X \rightarrow Y$ is continuous. Show that if $A \subset X$ is connected, then $f(A) \subset Y$ is connected.
5. Suppose that X and Y are metric spaces and that X is compact. Show that if $f : X \rightarrow Y$ is continuous, then f is uniformly continuous.

Examples and Minor Proofs.

In this section, give brief examples, counterexamples, or quick proofs.

6. State Urysohn's Lemma.
7. Is it possible for there to be a continuous function $f : [0, 1] \rightarrow \{0, 1\}$ which is onto?
8. State the Bolzano–Weierstrass Theorem.
9. Define a continuous function $f : C \rightarrow [0, 1]$ which is onto where C is the Cantor set and $[0, 1]$ is the unit interval.
10. Give an example of a space X which is T_1 but not Hausdorff.
11. State the Tietze Extension Theorem.
12. Define a quotient map.
13. State the Baire Category Theorem.
14. What does it mean for a topological space X to be Hausdorff? Regular? Normal?
15. What does it mean for a space X to be first countable? Show that every metric space is first-countable.